

[ARCHIVED CATALOGUE]

Electrical and Computer Engineering (Hardware Systems for Machine Learning) (MS)

In addition to the general requirements of the Viterbi School of Engineering, the Master of Science in Electrical and Computer Engineering with an emphasis in Machine Learning Systems is also subject to the following requirements.

A minimum of 32 units, selected from the courses below, is required. Units to be transferred (maximum 4 with adviser approval) must have been taken prior to taking classes at USC – interruption of residency is not allowed. Curricular Practical Training units do not count toward the 32 required units.

Recommended Preparation

A maximum of two of the following courses (8 units) may be counted toward the degree if needed to prepare for the 500-level courses.

- [CSCI 402 Operating Systems](#) Units: 4
- [EE 451 Parallel and Distributed Computation](#) Units: 4
- [EE 454L Introduction to System-on-Chip](#) Units: 4
- [EE 457 Computer Systems Organization](#) Units: 4
- [EE 477L MOS VLSI Circuit Design](#) Units: 4

Required Hardware Systems Courses

Take all three courses from the required list (12 units)

- [EE 508 Hardware Foundations of Machine Learning](#) Units: 4
- [EE 511 Machine Learning Hardware Accelerators](#) Units: 4
- [EE 560L Digital System Design](#) Units: 4

Electives

Data Processing

Take at least one course from this list:

- [CSCI 544 Applied Natural Language Processing](#) Units: 4
- [CSCI 677 Advanced Computer Vision](#) Units: 4
- [EE 519 Speech Recognition and Processing for Multimedia](#) Units: 3
- [EE 569 Introduction to Digital Image Processing](#) Units: 4
- [EE 669 Multimedia Data Compression](#) Units: 4

Mathematics and Architecture

Take three courses from the following lists:

Mathematics

- [EE 503 Probability for Electrical and Computer Engineers](#) Units: 4
- [EE 510 Linear Algebra for Engineering](#) Units: 4

Architecture and Design

- [EE 533 Network Processor Design and Programming](#) Units: 4
- [EE 557 Computer Systems Architecture](#) Units: 4
- [EE 577a VLSI System Design](#) Units: 4
- [EE 577b VLSI System Design](#) Units: 4
- [EE 580 System Verification](#) Units: 4

Additional Electives

Take as many courses as needed from the following list to complete the units for the degree:

- [EE 559 Machine Learning I: Supervised Methods](#) Units: 4
 - [EE 582 CMOS:Nano Neuromorphic Circuits](#) Units: 4
 - [EE 588 Optimization for the Information and Data Sciences](#) Units: 4
 - [EE 653 Advanced Topics in Microarchitecture](#) Units: 3
 - [EE 657 Parallel and Distributed Computing](#) Units: 3
 - [EE 658 Diagnosis and Design of Reliable Digital Systems](#) Units: 4
 - [EE 659 Advanced Topics in Interconnection Network Design and Analysis](#) Units: 4
 - [EE 660 Machine Learning II: Mathematical Foundations and Methods](#) Units: 4
 - [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2
 - [EE 680 Computer-Aided Design of Digital Systems](#) Units: 4
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